Georg Bökman

boekman@outlook.com
https://georg-bn.github.io
https://scholar.google.com/citations?user=FUE3Wd0AAAAJ

Timeline

2025–	**Postdoc in the Amsterdam Machine Learning Lab (AMLab)** University of Amsterdam, Amsterdam Supervised by Erik J. Bekkers
2024–2025	**Researcher in the Computer Vision group** Chalmers University of Technology, Gothenburg
2021–2024	**Leader of the Geometric Deep Learning cluster** A group of PhD students from universities around Sweden within the Wallenberg AI, Autonomous Systems and Software Program (WASP). Organized study trips to Amsterdam and Zürich/Lausanne, as well as research and study groups.
2019–2024	**PhD in the Computer Vision group** Chalmers University of Technology, Gothenburg Research on equivariant deep learning for computer vision applications Supervised by Fredrik Kahl
2016–2019	**Master of Science in Engineering mathematics** Chalmers University of Technology, Gothenburg
2016–2017	**Exchange year at the Master's programme in Mathematics** Technical University of Munich, Munich
2013–2016	**Bachelor of Science in Engineering mathematics** Chalmers University of Technology, Gothenburg

Selected publications

ECCV 2026	**Quick ViTs: Speeding up vision transformers through equivariance** David Nordström, Johan Edstedt, Fredrik Kahl, **Georg Bökman** arXiv:2505.15441
ICML 2026	**Identifiable equivariant networks are layerwise equivariant** Vahid Shahverdi[*], Giovanni Luca Marchetti[*], **Georg Bökman**[*], Kathlén Kohn[*] arXiv:2601.21645
ICML 2026	**Platonic transformers: A solid choice for equivariance** Mohammad Mohaiminul Islam, Rishabh Anand, David R. Wessels, Friso de Kruiff, Thijs P. Kuipers, Rex Ying, Clara I. Sánchez, Sharvaree Vadgama, **Georg Bökman**, Erik J. Bekkers arXiv:2510.03511
ICML 2025 Spotlight	**Flopping for FLOPs: Leveraging equivariance for computational efficiency** **Georg Bökman**, David Nordström, Fredrik Kahl arXiv:2502.05169
CVPR 2024 Oral	**Steerers: A framework for rotation equivariant keypoint descriptors** **Georg Bökman**, Johan Edstedt, Michael Felsberg, Fredrik Kahl arXiv:2312.02152

Awards/Achievements

March 2026	Most Industry Relevant Paper, Swedish Symposium on Image Analysis, Örebro
March 2024	Most Industry Relevant Paper, Swedish Symposium on Image Analysis, Luleå
August 2023	Winner AISG/SLA Visual Localization Challenge ($20 000), IJCAI, Macao
January 2023	Best Poster Award, WASP Winter Conference, Norrköping

[*] Equal contribution

Mentees

2026–	Tīkun Ông Research mentee
2024–	David Nordström PhD student, Chalmers (co-supervisor with Fredrik Kahl as main supervisor)

Academic service

Area Chair	ICLR 2027
Reviewing	Regularly at ICML, ICLR, NeurIPS, CVPR, ECCV, TMLR Outstanding/Top reviewer at NeurIPS 2024, NLDL 2026, ICLR 2026, CVPR 2026, ECCV 2026

Teaching

2019–2023	Teaching assistant for a master's level course in computer vision (Chalmers).
2017–2023	Teaching assistant for various undergraduate courses in mathematics, including calculus, linear algebra, real analysis, and computational mathematics (Chalmers).

Further publications

NeurReps 2026	**A unified framework for vision transformers equivariant to discrete subgroups of O(2)** Tīkun Ông, **Georg Bökman** arXiv:2606.27864
ECCV 2026 Spotlight oral	**LoMa: Local feature matching revisited** David Nordström*, Johan Edstedt*, **Georg Bökman**, Jonathan Astermark, Anders Heyden, Viktor Larsson, Mårten Wadenbäck, Michael Felsberg, Fredrik Kahl arXiv:2604.04931
ECCV 2026 Spotlight oral	**RoMa v2: Harder better faster denser feature matching** Johan Edstedt, David Nordström, Yushan Zhang, **Georg Bökman**, Jonathan Astermark, Viktor Larsson, Anders Heyden, Fredrik Kahl, Mårten Wadenbäck, Michael Felsberg arXiv:2511.15706
CVPR 2026	**MuM: Multi-view masked image modeling for 3D vision** David Nordström, Johan Edstedt, Fredrik Kahl, **Georg Bökman** arXiv:2511.17309
IMW 2026	**Who handles orientation? Investigating invariance in feature matching** David Nordström, Johan Edstedt, Fredrik Kahl, **Georg Bökman** arXiv:2604.11809
3DV 2026	**A framework for reducing the complexity of geometric vision problems and its application to two-view triangulation with approximation bounds** Felix Rydell, **Georg Bökman**, Fredrik Kahl, Kathlén Kohn arXiv:2503.08142
ECCV 2024	**Affine steerers for structured keypoint description** **Georg Bökman**, Johan Edstedt, Michael Felsberg, Fredrik Kahl arXiv:2408.14186
CVPR 2024	**RoMa: Robust dense feature matching** Johan Edstedt, Qiyu Sun, **Georg Bökman**, Mårten Wadenbäck, Michael Felsberg arXiv:2305.15404
IMW 2024	**DeDoDe v2: Analyzing and improving the DeDoDe keypoint detector** Johan Edstedt, **Georg Bökman**, Zhenjun Zhao arXiv:2404.08928
3DV 2024 Oral	**DeDoDe: Detect, don't describe—describe don't detect for local feature matching** Johan Edstedt, **Georg Bökman**, Mårten Wadenbäck, Michael Felsberg arXiv:2308.08479
TMLR 2024	**In search of projectively equivariant networks** **Georg Bökman***, Axel Flinth*, Fredrik Kahl arXiv:2209.14719
NeurIPS 2023	**Investigating how ReLU-networks encode symmetries** **Georg Bökman**, Fredrik Kahl arXiv:2305.17017

SCIA 2023	**Rigidity preserving image transformations and equivariance in perspective** Lucas Brynte*, **Georg Bökman***, Axel Flinth, Fredrik Kahl arXiv:2201.13065
CVPR 2022 Oral	**ZZ-Net: A universal rotation equivariant architecture for 2D point clouds** **Georg Bökman**, Fredrik Kahl, Axel Flinth arXiv:2111.15341
IMW 2022	**A case for using rotation invariant features in state of the art feature matchers** **Georg Bökman**, Fredrik Kahl arXiv:2204.10144
ICPR 2022 Oral	**Azimuthal rotational equivariance in spherical CNNs** Carl Toft, **Georg Bökman**, Fredrik Kahl

Languages

Swedish:	Native
German:	Native
English:	Fluent
Spanish:	Basic

Hobbies

I enjoy choral singing and playing the piano and guitar in my free time.

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